

Grammar is said to be Ambiguous if
 there exists more than one

(i) LMD or (ii) RMD or (iii) Parse tree

for a same Input String.

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * E \mid E$$

$$F \rightarrow id$$

$$1) E \rightarrow E + E \mid E - E \mid id$$

$$id + id - id$$

$$E \rightarrow E + E$$

$$\rightarrow id + E$$

$$\rightarrow id + E - E$$

$$\rightarrow id + id - E$$

$$\rightarrow id + id - id$$

$$E \rightarrow E - E$$

$$\rightarrow id - [E + E - E]$$

$$\rightarrow id + E - E$$

$$\rightarrow id + id - E$$

$$\rightarrow id + id - id$$

For same Input String, we Constructed two parse
 tree / derivation. So it is Ambiguous.

$$2) S \rightarrow S + S \mid S * S \mid a \mid b \quad a + a * b$$

$$3) S \rightarrow a S b S \mid b S a S \mid \epsilon \quad abab.$$

Unambiguous Grammar

Does not contain more than one LMD, RMD, Parse
 tree for Given String.

1) If the left associative Operators (+, -, *, /) are used
 in production rule, then apply left recursion
in the production Rule.

* Left recursion means left most Symbol on the
 right side is same as non-terminal on the
 left side.

$$Ex: \underline{X} \rightarrow \underline{X} a$$

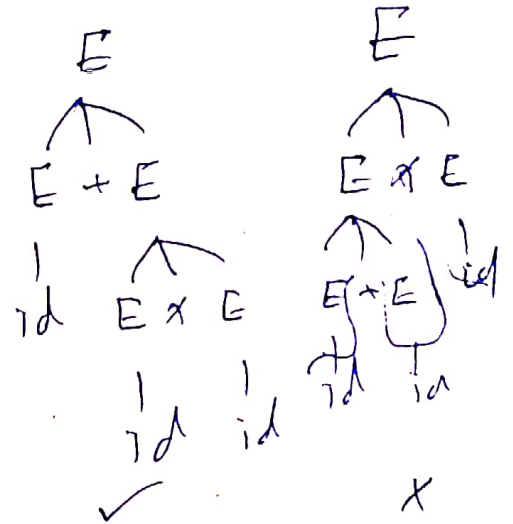
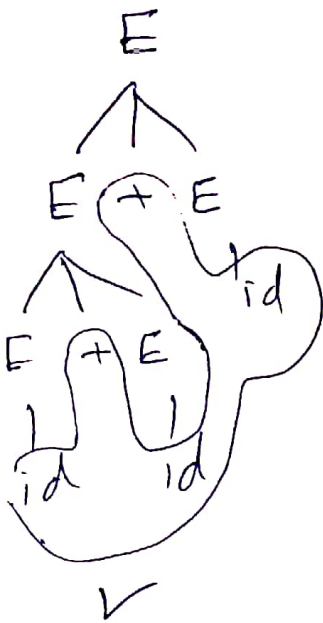
Ambiguous Removal

$$E \rightarrow E + E \mid E * E \mid id$$

id + id + id

id
+

② id + id * id
①



X

$$E \rightarrow E + id \mid id$$

$$E \rightarrow E + T$$

$$T \rightarrow id$$

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow id$$

(I) By adding Precedence

(II) By adding associativity: (Left Recursive + left association)
Right " Right "

id
*
+
↑ Bottom-up Approach

→ Add precedence levels.

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow id$$

→ Add variables to represent each precedence level

R

11.

CFG $\rightarrow$ CNF

$S \rightarrow aA \mid aBB$

$A \rightarrow aAA \mid \epsilon$

$B \rightarrow bB \mid bbC$

$C \rightarrow B$

(i) Introduce new variable for all terminal (length ≥ 2)

$P \rightarrow a$

(ii) Break the productions with length $3 \Rightarrow 2$

If body $\Rightarrow K$

$K-2$ Cascade Symbols

(i) Eliminate ϵ productions.

Nullable Symbols: $\{A\}$

$S \rightarrow aA \mid aBB \mid a$

$A \rightarrow aAA \mid aA \mid a$

$B \rightarrow bB \mid bbC$

$C \rightarrow B$

Ex: $S \rightarrow ABCD$

4
 $4-2=2$

$S \rightarrow AC_1$

$C_1 \rightarrow BC_2$

$C_2 \rightarrow CD$

(ii) Eliminate unit productions.

$S \rightarrow aA \mid aBB \mid a$

$A \rightarrow aAA \mid aA \mid a$

$B \rightarrow bB \mid bbC$

$C \rightarrow bB \mid bbC$

(iii) Eliminate Useless Symbols

Generating Symbols: $\{a, b, S, A\}$

Non-Generating Symbols: $\{B, C\}$

$S \rightarrow aA \mid a$

$A \rightarrow aAA \mid aA \mid a$

Reachable Symbols: $\{S, a, A\}$

(IV) New Variable for Terminal

$$P \rightarrow a$$

If length ≥ 2 , Replace a with P .

$$P \rightarrow a$$

$$S \rightarrow PA \mid a$$

$$A \rightarrow PAA \mid PA \mid a$$

{a,b}

(V) CNF Break production body if length > 2 $3-2=1$

$$\begin{array}{l} P \rightarrow a \\ S \rightarrow PA \mid a \\ A \rightarrow PC_1 \mid PA \mid a \\ C_1 \rightarrow AA \end{array}$$

$$2) S \rightarrow ABC$$

$$A \rightarrow BC \mid a$$

$$B \rightarrow bAC \mid \epsilon$$

$$C \rightarrow cAB \mid \epsilon$$

(i) Eliminate ϵ production

$$S \rightarrow ABC \mid AC$$

$$A \rightarrow BC \mid \epsilon \mid a$$

$$B \rightarrow bAC$$

$$C \rightarrow cAB \mid cA \mid \epsilon$$

$$S \rightarrow ABC \mid BC \mid AC \mid C \mid AB \mid B \mid A$$

$$A \rightarrow BC \mid B \mid C \mid a$$

$$B \rightarrow bAC \mid bC \mid bA \mid b$$

$$C \rightarrow cAB \mid cB \mid cA \mid c$$

$$S \rightarrow ABC \mid AC \mid AB \mid A$$

$$A \rightarrow BC \mid B \mid C \mid \epsilon \mid a$$

$$B \rightarrow bAC \mid bA$$

$$C \rightarrow cAB \mid cA$$

minimize unlike production

$\rightarrow ABC | BC | AC | \underline{C} | \underline{AB} | \underline{B} | \underline{A}$

$\rightarrow BC | \underline{B} | C | a$

$\rightarrow bAC | bC | bA | b$

$\rightarrow CAB | cB | cA | c$

minimize $S \rightarrow \underline{BA}$

$S \rightarrow ABC | BC | AC | C | AB | \underline{B} | a$

$A \rightarrow BC | B | C | a$

~~$B \rightarrow bAC$~~

eliminate $S \rightarrow B$

$S \rightarrow ABC | BC | AC | C | AB | bAC | bC | bA | b | a$

eliminate $S \rightarrow C$

$S \rightarrow ABC | BC | AC | CAB | CB | cA | c | AB | bAC | bC | bA | b | a$ (13)

eliminate $A \rightarrow B$

$A \rightarrow BC | bAC | bC | bA | b | C | a$

eliminate $A \rightarrow C$

$A \rightarrow BC | bAC | bC | bA | b | CAB | CB | cA | c | a$ (10)

eliminate unlike symbol

$S \rightarrow ABC | BC | AC | CAB | CB | cA | c | AB | bAC | bC | bA | b | a$

$A \rightarrow BC | bAC | bC | bA | b | CAB | CB | cA | c | a$

$B \rightarrow bAC | bC | bA | b$

$C \rightarrow CAB | CB | cA | c$

Generating Symbols = $\{S, A, B, C, a, b, c\}$ Non-Generating = $\emptyset$

Reachable Symbols = $\{S, A, B, C, a, b, c\}$ Non-Generating = $\emptyset$

(iv) Transforming Production where $|body| > 1$

$$S \rightarrow \overset{\checkmark}{ABC} | \overset{\checkmark}{BC} | \overset{\checkmark}{AC} | \overset{\checkmark}{EAB} | \overset{\checkmark}{EB} | \overset{\checkmark}{EA} | \overset{\checkmark}{e} | \overset{\checkmark}{AB} | \overset{\checkmark}{DAC} | \overset{\checkmark}{DC} | \overset{\checkmark}{DA} | \overset{\checkmark}{b} | \overset{\checkmark}{c}$$

$$A \rightarrow \overset{\checkmark}{BC} | \overset{\checkmark}{DAC} | \overset{\checkmark}{DC} | \overset{\checkmark}{DA} | \overset{\checkmark}{b} | \overset{\checkmark}{EAB} | \overset{\checkmark}{EB} | \overset{\checkmark}{EA} | \overset{\checkmark}{e} | \overset{\checkmark}{a}$$

$$B \rightarrow \overset{\checkmark}{DAC} | \overset{\checkmark}{DC} | \overset{\checkmark}{DA} | \overset{\checkmark}{b}$$

$$C \rightarrow \overset{\checkmark}{EAB} | \overset{\checkmark}{EB} | \overset{\checkmark}{EA} | \overset{\checkmark}{e}$$

$$D \rightarrow b$$

$$E \rightarrow c$$

(v) Breaking Production where $|body| > 2$

$$S \rightarrow EAB$$

$$B \rightarrow DAC$$

$$C \rightarrow EAB$$

$$S \rightarrow EC_1$$

$$B \rightarrow DC_2$$

$$C \rightarrow EC_3$$

$$C_1 \rightarrow AB$$

$$C_2 \rightarrow AC$$

$$C_3 \rightarrow AB$$

3) $S \rightarrow a | aA | B$

$$A \rightarrow aBB | \epsilon$$

$$B \rightarrow Aa | b$$

1) Eliminate $A \rightarrow \epsilon$

Eliminate $S \rightarrow B$

$$S \rightarrow a | aA | B$$

$$S \rightarrow a | aA | Aa | b$$

$$A \rightarrow aBB$$

$$A \rightarrow aBB$$

$$B \rightarrow Aa | a | b$$

$$B \rightarrow Aa | a | b$$

No union Symbol.

$|body| > 1$

$$S \rightarrow \check{a} | \check{CA} | \check{AC} | b$$

$$A \rightarrow CBB$$

$$B \rightarrow A\check{C} | \check{a} | b$$

$|body| > 2$

$$A \rightarrow CBB$$

$$A \rightarrow CC_1$$

$$C_1 \rightarrow BB$$

$$S \rightarrow a | CA | AC | b$$

$$A \rightarrow CC_1$$

$$B \rightarrow AC | a | b$$

$$C_1 \rightarrow BB$$

$$C \rightarrow a$$

Back - Normal Form

$$A \rightarrow a\alpha$$

$A \Rightarrow$ Variable

$a \Rightarrow$ Terminal

α = Zero or more Variables.

Procedure for Converting CFG to GNF

Step 1: Check whether given CFG is in CNF.

If not, Convert into CNF

Step 2: Replace all variables of CFG as $A_1, A_2, \dots, A_n$.

$A_i \Rightarrow$ Start Symbol

Step 3: $A_i \rightarrow A_j \beta$

(i) $i < j$

(ii) $i > j$

(iii) $i = j$

Case i: $i < j$

If $A_i \rightarrow A_j \beta$ & $i < j$

Leave production as it is.

Ex: $A_1 \rightarrow A_2 a$

Case ii: $i > j$

If $A_i \rightarrow A_j \beta$ & $i > j$

Replace A_j with its production bodies, such that

If $A_j \rightarrow \alpha_1 | \alpha_2 | \dots | \alpha_n$

Then $A_i \rightarrow \alpha_1 \beta | \alpha_2 \beta | \dots | \alpha_n \beta$.

Case iii $i = j$

If $A_i \rightarrow A_j B \mid \alpha$ $i = j$

Introduce new Variable B .

$$B \rightarrow \beta \mid \beta B$$

$$A_i \rightarrow \alpha \mid \alpha B$$

Step 4: Repeat Step 3 until we reach the productions of the form $A \rightarrow a \alpha$

Example:

Convert the following Grammar into GNF

$$S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

Step 1: The given CFG is in CNF

Step 2: Renaming

$$S = A_1 \quad A = A_2$$

$$A_1 \rightarrow A_2 A_2 \mid 0$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

Step 3: (i) $A_1 \rightarrow A_2 A_2 \mid 0 \rightarrow \textcircled{1}$

$$i = 1 \quad j = 2 \quad (i < j)$$

$$(ii) A_2 \rightarrow A_1 A_1 \mid 1$$

$$i = 2 \quad j = 1 \quad (i > j) \text{ Convert}$$

$$A_2 \rightarrow A_2 A_2 A_1 \mid A_0 A_1 \mid 1$$

$$i = 2 \quad j = 2 \quad (i = j) \text{ Case iii}$$

$A_2 A_2 A_1$
 $\downarrow$
Left Recursive
Production

1. Introduce new Variable B_2

$$B_2 \rightarrow A_2 A_1 \mid A_2 A_1 B_2 \rightarrow (2)$$

$$A_2 \rightarrow 0A_1 \mid 1 \mid 0A_1 B_2 \mid 1B_2 \rightarrow (3)$$

(i) $A_1 \rightarrow A_2 A_2 \mid 0$ Sub (3) in (1)

$$A_1 \rightarrow 0A_1 A_2 \mid 1A_2 \mid 0A_1 B_2 A_2 \mid 1B_2 A_2 A_2 \mid 0 \rightarrow (4)$$

(ii) $B_2 \rightarrow A_2 A_1 \mid A_2 A_1 B_2$ Sub (3) in (2)

$$B_2 \rightarrow 0A_1 A_1 \mid 1A_1 \mid 0A_1 B_2 A_1 \mid 1B_2 A_1 \mid$$

$$0A_1 A_1 B_2 \mid 1A_1 B_2 \mid 0A_1 B_2 A_1 B_2 \mid 1B_2 A_1 B_2$$

$$\therefore \dots \rightarrow (5)$$

Now A_1, A_2, B_2 are in GNF format

GNF

$$A_1 \rightarrow 0A_1 A_2 \mid 1A_2 \mid 0A_1 B_2 A_2 \mid 1B_2 A_2 A_2 \mid 0$$

$$A_2 \rightarrow 0A_1 \mid 1 \mid 0A_1 B_2 \mid 1B_2$$

$$B_2 \rightarrow 0A_1 A_1 \mid 1A_1 \mid 0A_1 B_2 A_1 \mid 1B_2 A_1 \mid 0A_1 A_1 B_2 \mid 1A_1 B_2 \mid 0A_1 B_2 A_1 B_2 \mid 1B_2 A_1 B_2$$

2) CFG to GNF

$$S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow CA$$

$$C \rightarrow AB \mid b$$

Step 1: Already in CNF

Step 2: Renaming $A_1, A_2, \dots$

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow a$$

$$A_3 \rightarrow A_4 A_2$$

$$A_4 \rightarrow A_2 A_3 \mid b$$

Step 3:

$$A_1 \rightarrow A_2 A_3 \quad (i < j) \rightarrow (1)$$

$$A_2 \rightarrow a \rightarrow (2)$$

$$A_3 \rightarrow A_4 A_2 \quad (i < j), \text{ (3)} \rightarrow (3)$$

$$A_4 \rightarrow A_2 A_3 \mid b \quad (i > j) \rightarrow (4)$$

$$A_4 \rightarrow a A_3 \mid b \rightarrow (4)$$

(2) & (4) in GNF

Take equation (1) $\Rightarrow$ Sub (2) in (1)

$$A_1 \rightarrow A_2 A_3$$

$$A_1 \rightarrow a A_3 \rightarrow (5)$$

Take equation (3) $\Rightarrow$ Sub (4) in (3)

$$A_3 \rightarrow A_4 A_2$$

$$A_3 \rightarrow a A_3 A_2 \mid b A_2 \rightarrow (6)$$

$$A_1 \rightarrow A_2 A_3 \mid b$$

$\Rightarrow$

$$A_3 \rightarrow A_4 A_2$$

$$\rightarrow a A_3 A_2 \mid b A_2$$

$\rightarrow (6)$

$$A_1 \rightarrow a A_3$$

$$A_2 \rightarrow a$$

$$A_3 \rightarrow a A_3 A_2 \mid b A_2$$

$$A_4 \rightarrow a A_3 \mid b$$

$$S \rightarrow aABb \mid bBaA$$

$$A \rightarrow cA \mid \epsilon c$$

$$B \rightarrow \epsilon \mid dB$$

Step 1: Eliminating ϵ production

$$S \rightarrow aABb \mid aAb \mid bBaA \mid baA$$

$$A \rightarrow cA \mid c$$

$$B \rightarrow dB \mid d$$

No unit production, No useless symbols.

Transforming variable (body) > 1

$$S \rightarrow CABb \mid CAB \mid bBaA \mid baA$$

$$A \rightarrow \epsilon A \mid c$$

$$B \rightarrow \epsilon DB \mid d$$

$$C \rightarrow c$$

$$D \rightarrow d$$

$$E \rightarrow b$$

$$F \rightarrow a$$

$$S \rightarrow CABb \mid CAE \mid EBFA \mid EFA$$

$$A \rightarrow CA \mid c$$

$$B \rightarrow DB \mid d$$

$$C \rightarrow c \quad D \rightarrow d \quad E \rightarrow b \quad F \rightarrow a$$

$$S \rightarrow CAEB$$

$$S \rightarrow CX_1$$

$$X_1 \rightarrow AEB$$

$$X_1 \rightarrow AX_2$$

$$X_2 \rightarrow EB$$

$$S \rightarrow CAE$$

$$S \rightarrow CX_3$$

$$X_3 \rightarrow AE$$

$$S \rightarrow EFA$$

$$S \rightarrow EX_4$$

$$X_4 \rightarrow PA$$

$$S \rightarrow EBFA$$

$$\rightarrow X_2FA$$

$$S \rightarrow X_2X_4$$

S C X₁ X₃ X₄ A B D E F X₂
A₁ A₂ A₃ A₄ A₅ A₆ A₇ A₈ A₉ A₁₀ A₁₁ A₁₂

$$S \rightarrow CX_1 \mid CX_3 \mid X_2X_4$$

$$A \rightarrow CA \mid c$$

$$B \rightarrow DB \mid d$$

$$C \rightarrow c$$

$$D \rightarrow d$$

$$E \rightarrow b$$

$$F \rightarrow a$$

$$X_1 \rightarrow AX_2$$

$$X_2 \rightarrow EB$$

$$X_3 \rightarrow AE$$

$$X_4 \rightarrow FA$$

$$SA_1 \rightarrow A_2A_3 \mid A_2A_4 \mid A_{11}A_5$$

$$A_6 \rightarrow A_2A_6 \mid c$$

$$A_7 \rightarrow A_8A_7 \mid d$$

$$A_2 \rightarrow c$$

$$A_8 \rightarrow d$$

$$A_9 \rightarrow b$$

$$A_{10} \rightarrow a$$

$$A_3 \rightarrow A_6A_{11}$$

$$A_{11} \rightarrow A_9A_7$$

$$A_4 \rightarrow A_6A_8$$

$$A_5 \rightarrow A_{10}A_6$$

$$A_1 \rightarrow A_2A_3 \mid A_2A_4 \mid A_{11}A_5$$

$$A_6 \rightarrow A_2A_6 \mid c \Rightarrow A_6 \rightarrow cA_6 \mid c \rightarrow \textcircled{1}$$

$$A_{11} \rightarrow A_9A_7 \rightarrow A_{11} \rightarrow bA_7 \rightarrow \textcircled{2}$$

$$A_1 \rightarrow cA_3 \mid cA_4 \mid bA_7A_5 \rightarrow \textcircled{3}$$

$$A_7 \rightarrow dA_7 \mid d \rightarrow \textcircled{4}$$

$$A_3 \rightarrow cA_6A_{11} \mid cA_{11} \rightarrow \textcircled{5}$$

$$A_4 \rightarrow cA_6A_8 \mid cA_8 \rightarrow \textcircled{6}$$

$$A_5 \rightarrow A_{10}A_6 \rightarrow A_5 \rightarrow aA_6 \rightarrow \textcircled{7}$$

1) Construct NPDA for $L = \{ww^R \mid w \text{ is in } (0+1)^+\}$

PDA for this Language L is $|w|/2, 0$

$P = (\{q_0, q_1, q_2\}, \{0, 1\}, (z_0, 0, 1), \delta, q_0, z_0, \{q_2\})$

Step 1: Initial Symbols in w are pushed into stack using start state q_0

$$\delta(q_0, 0, z_0) = (q_0, 0z_0)$$

$$\delta(q_0, 1, z_0) = (q_0, 1z_0)$$

$$\delta(q_0, 0, 0) = (q_0, 00)$$

$$\delta(q_0, 1, 0) = (q_0, 10)$$

$$\delta(q_0, 0, 1) = (q_0, 01)$$

$$\delta(q_0, 1, 1) = (q_0, 11)$$

Step 2: Transition moving from q_0 to q_1 by processing ϵ . No changes in stack

$$\delta(q_0, \epsilon, 0) = (q_1, 0)$$

$$\delta(q_0, \epsilon, 1) = (q_1, 1)$$

$$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$$

Step 3: Verifying Symbol in stack top with corresponding symbol in w^R and perform pop

$$\delta(q_1, 0, 0) = (q_1, \epsilon)$$

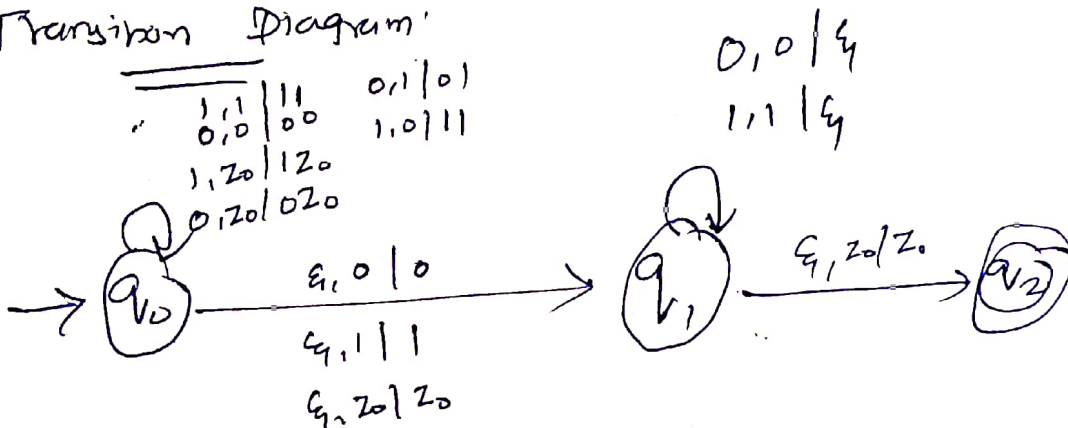
$$\delta(q_1, 1, 1) = (q_1, \epsilon)$$

Step 4: Transition accepts the string by processing ϵ and verifying stack top is having only Z_0 .

$$\delta(q_1, \epsilon, Z_0) = (q_2, Z_0)$$

$$\delta(q_1, \epsilon, Z_0) = (q_2, \epsilon)$$

Transition Diagram:



$1, 0 | 1 \Rightarrow$ Remove $\epsilon, Z_0 | Z_0$

2) $L = 0^n 1^n \mid n \geq 0$

Transitions:

- $q_0 \rightarrow q_0$ (self-loop): $0, 0 | 00$, $0, Z_0 | 0Z_0$
- $q_0 \rightarrow q_1$: $1, 0 | \epsilon$, $\epsilon, Z_0 | Z_0$
- $q_1 \rightarrow q_1$ (self-loop): $1, 0 | \epsilon$
- $q_1 \rightarrow q_2$: $\epsilon, Z_0 | Z_0$

3) $L = 0^n 1^{2n} \mid n \geq 1$

Transitions:

- $q_0 \rightarrow q_0$ (self-loop): $0, 0 | 00$, $0, Z_0 | 0Z_0$
- $q_0 \rightarrow q_1$: $1, 0 | 0$
- $q_1 \rightarrow q_1$ (self-loop): $1, 0 | \epsilon$
- $q_1 \rightarrow q_2$: $1, 0 | \epsilon$
- $q_2 \rightarrow q_1$: $1, 0 | 0$
- $q_2 \rightarrow q_3$: $\epsilon, Z_0 | Z_0$

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$$

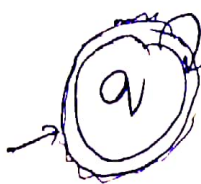
By Final State

$$L(M) = \{w \in \Sigma^* \mid (q_0, w, z_0) \vdash^* (p, \epsilon, \epsilon) \text{ for } p \in F\}$$

By Empty Stack

$$N(M) = \{w \in \Sigma^* \mid (q_0, w, z_0) \vdash^* (p, \epsilon, \epsilon) \text{ for any } p \in Q\}$$

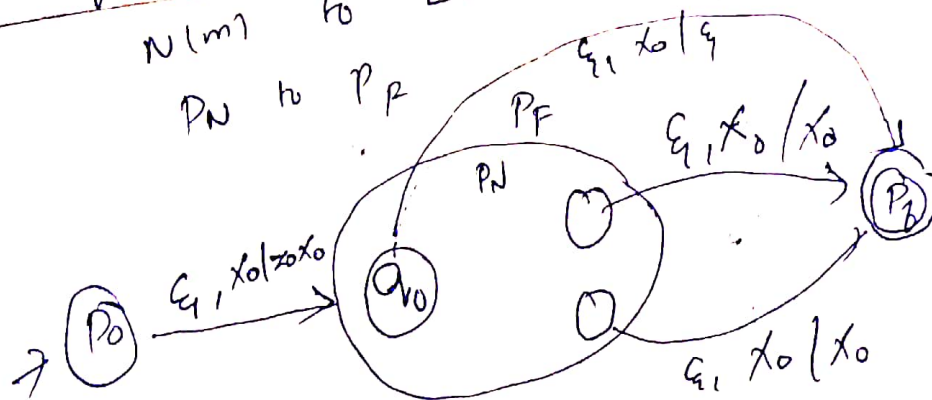
Ex: $0^n 1^n \mid n \geq 0$



0, z0 | 0 z0
0 1 0 | 0 0
1 1 0 | ϵ
 ϵ , z0 | ϵ

$(\{q\}, \{0, 1\}, \{0, z_0\}, \delta, q, z_0, \emptyset)$

Empty Stack to Final State Acceptance.
 $N(M)$ to $L(M)$
 P_N to P_F
 $P_N = (Q, \Sigma, \Gamma, \delta_N, q_0, z_0)$



$$P_F = (Q \cup \{p_0, p_1\}, \Sigma, \Gamma \cup \{x_0\}, \delta_F, p_0, \{p_1\})$$

$$\delta(p_0, q_1, x_0) = (q_0, z_0 x_0)$$

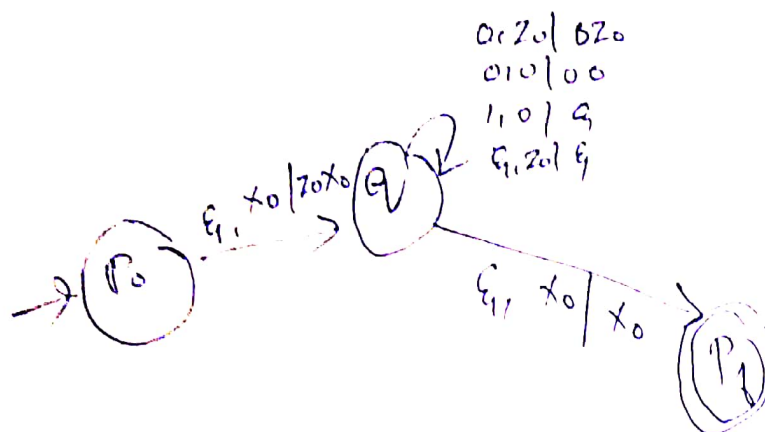
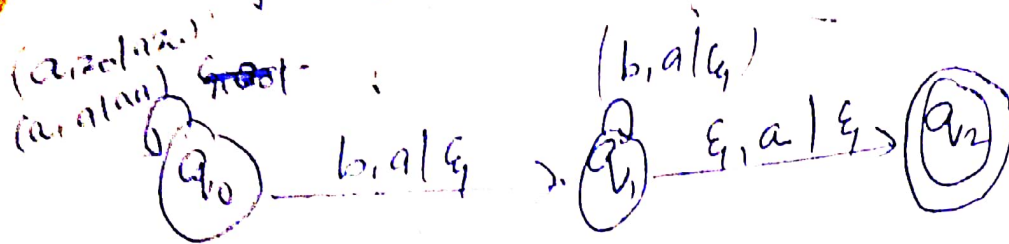
$$\text{for each } a \text{ in } Q, x \text{ in } \Gamma, a \neq q_1 \text{ in } \Sigma$$

$$\delta_r(a, a, x) = \emptyset \text{ for } \delta(a, a, x)$$

$$\delta_r(a, q_1, x_0) = (p_1, x_0)$$

PDA for $L = \{a^n b^m \mid n \geq 1, m \geq 1\}$
 atleast 1 b's, 2 a's

$L = \{aab, aaabb, aaaaabbb\}$



Equivalent to

$a^n z_0 b^m$

$a^n z_0 b^m$

(1)

$a^n z_0 b^m$

$a^n z_0 b^m$

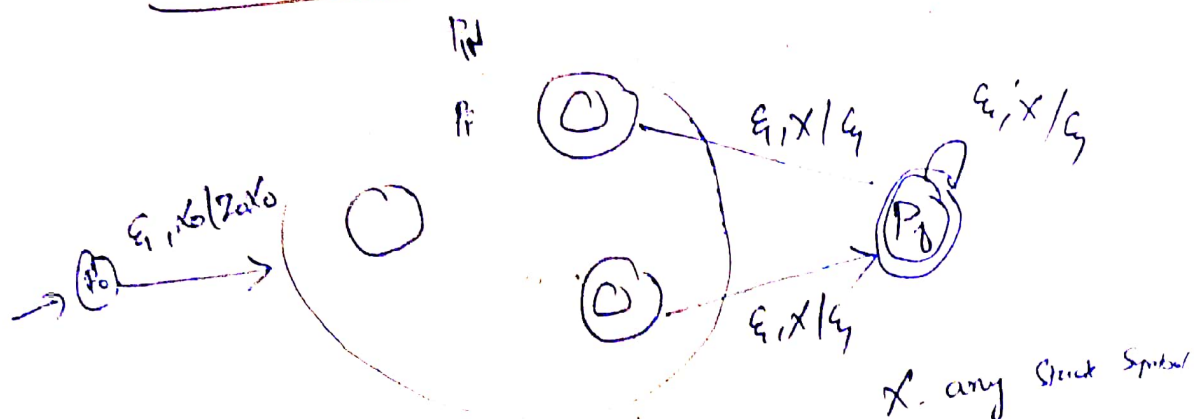
$a^n z_0 b^m$

$a^n z_0 b^m$

$a^n z_0 b^m$

$a^n z_0 b^m$

From Final State Acceptance to Empty Stack Acceptance



$G = (V, T, P, S)$

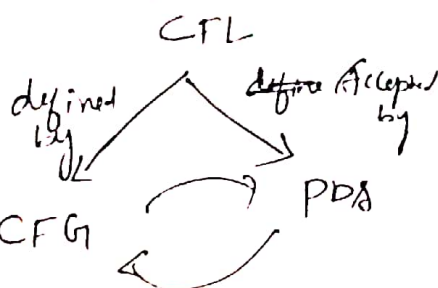
CFG to PDA



$N(P) \rightarrow PDA$ by Empty Stack

$P = (\{q\}, T, V \cup T, \delta, q, S)$

$P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0)$



Rule 1: For each Variable A

$\delta(q, \epsilon, A) = (q, B)$ where $A \rightarrow B$ is production

Rule 2: For each terminal 'a'

$\delta(q, a, a) = (q, \epsilon)$

Ex: $S \rightarrow OS1 | 0011$

The equivalent PDA for given grammar.

$\delta(q, \epsilon, S) = \{(q, OS1), (q, 00), (q, 11)\} \rightarrow \text{to } \text{0111}$

$\delta(q, 0, 0) = (q, \epsilon) \rightarrow \text{②}$

$\delta(q, 1, 1) = (q, \epsilon) \rightarrow \text{③}$

String 0111

$\delta(q, 0111, S) \rightarrow \text{using } \text{①}$

$\delta(q, 0111, OS1) \rightarrow \text{②}$

$\delta(q, 111, S1) \rightarrow \text{①}$

$\delta(q, 111, 111) \rightarrow \text{③}$

$\delta(q, 11, 11) \rightarrow \text{③}$

$\delta(q, 1, 1) \rightarrow \text{③}$

$\delta(q, \epsilon, \epsilon) \rightarrow \text{Accepted}$

$\delta(q, 0111, S) \vdash (q, 0111, OS1)$
 $\vdash (q, 111, S1)$
 $\vdash (q, 11, 111)$
 $\vdash (q, 1, 111)$
 $\vdash (q, 1, 1)$
 $\vdash (q, \epsilon, \epsilon)$

Accepted

2) Construct PDA for the following string
 whether $abbabb$ is in $N(P)$

$$G = (\{S, A\}, \{a, b\}, P, S)$$

$$P = \{ S \rightarrow AA \mid a$$

$$A \rightarrow SA \mid b \}$$

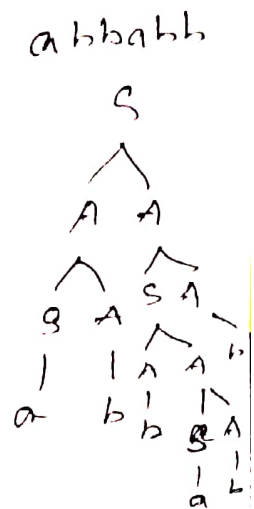
$$PDA, P = (\{q\}, \{a, b\}, \{S, A, a, b\}, \delta, q, S)$$

$$\delta(q, \epsilon, S) = \{(q, AA), (a, a)\}$$

$$\delta(q, \epsilon, A) = \{(q, SA), (a, b)\}$$

$$\delta(q, a, a) = (q, \epsilon)$$

$$\delta(q, b, b) = (q, \epsilon)$$



$$(q, abbabb, S) \vdash (q, abbabb, AA)$$

$$\vdash (q, abbabb, SA)$$

$$\vdash (q, abbabb, aAA)$$

$$\vdash (q, bbabb, AA)$$

$$\vdash (q, bbabb, bA)$$

$$\vdash (q, babb, A)$$

$$\vdash (q, babb, SA)$$

$$\vdash (q, babb, AAA)$$

$$\vdash (q, babb, bAA)$$

$$\vdash (q, abb, AA)$$

$$\vdash (q, abb, SAA)$$

$$\vdash (q, abb, aAA)$$

$$\vdash (q, bb, AA)$$

$$\vdash (q, bb, bA)$$

$$\vdash (q, b, A)$$

$$\vdash (q, b, b)$$

$$\vdash (q, \epsilon, \epsilon)$$

PDA to CFG

Let PDA, $P = (Q, \Sigma, \Gamma, \delta, q_0, z_0)$

CFG: $(V, \underline{\Sigma}, P, S)$

Variables, V

$S \rightarrow$ Start Symbol

1. Special Symbol S

2. $[pXa]$ where p, a are states in Q & X is in Γ

Productions, P

p in Q

$Q \times \Gamma \times Q$

$Q \times \Sigma \times Q$

1. For all states p ,

$$S \rightarrow [q_0 z_0 p]$$

2. Let $\delta(q, a, X) = (\gamma, Y_1 Y_2 \dots Y_k)$

$$(i) [q X Y_k] \rightarrow a [\gamma Y_1 \gamma_1] [\gamma_1 Y_2 \gamma_2] \dots [\gamma_{k-1} Y_k \gamma_k]$$

For all states $\gamma_1 \gamma_2 \dots \gamma_k$

$$(ii) \delta(q, a, X) = (\gamma, \epsilon)$$

$$[q X \gamma] \Rightarrow a$$

$$(iii) \delta(q, \epsilon, X) = (\gamma, \epsilon)$$

$$[q X \epsilon] \rightarrow \epsilon$$

1) Convert the following prod into

$$P = (\{a, p\}, \{0, 1\}, \{x, z\}, \delta, q, z)$$

$$\delta(q, 1, z) = (q, xz)$$

$$\delta(q, 1, x) = (q, xx)$$

$$\delta(q, 0, x) = (p, x)$$

$$\delta(p, 1, x) = (p, x)$$

$$\delta(p, 0, z) = (q, z)$$

Variables:

$$V = \{S, [a \times a], [a \times p], [p \times a], [p \times p], [a \times z], [a \times p], [p \times a], [p \times p]\}$$

$$S \rightarrow [a \times a]$$

$$S \rightarrow [a \times p]$$

(ii) $\delta(q, 1, z) = (q, xz)$

$$[a \times a] \rightarrow 1 [a \times a] [a \times a]$$

$$[a \times a] \rightarrow 1 [a \times p] [p \times a]$$

$$[a \times p] \rightarrow 1 [a \times a] [a \times p]$$

$$[a \times p] \rightarrow 1 [a \times p] [p \times p]$$

(iii) $\delta(q, 1, x) = (q, xx)$

$$[a \times a] \rightarrow 1 [a \times a] [a \times a]$$

$$[a \times a] \rightarrow 1 [a \times p] [p \times a]$$

$$[a \times p] \rightarrow 1 [a \times a] [a \times p]$$

$$[a \times p] \rightarrow 1 [a \times p] [p \times p]$$

$$g(q, \xi, x) = (q, \xi)$$

$$[q \times q] \rightarrow \xi$$

$$(v) \quad g(q, 0, x) = (p, x)$$

$$[q \times p] \rightarrow 0 [p \times p]$$

$$[q \times q] \rightarrow 0 [p \times q]$$

$$(vi) \quad g(p, 1, x) = (p, \xi)$$

$$[p \times p] \rightarrow 1$$

$$(vii) \quad g(p, 0, z) = (q, z)$$

$$[p \times q] \rightarrow 0 [q \times q]$$

$$[p \times p] \rightarrow 0 [q \times p]$$

$$2) \quad \{a_0, a_1\}, \{a, b\}, \{a, z_0\}, g, a_0, z_0, \phi$$

$$g(a_0, a, z_0) = (a_0, a z_0)$$

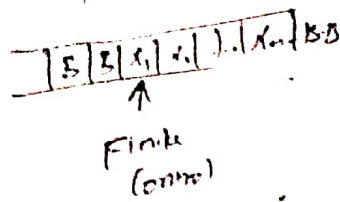
$$[a_0 \ z_0 \ a_0] \rightarrow a [a_0 \ a \ a_0] [a_0 \ z_0 \ a_0]$$

$$[a_0 \ z_0 \ a_0] \rightarrow a [a_0 \ a \ a_1] [a_0 \ z_0 \ a_0]$$

$$[a_0 \ z_0 \ a_1] \rightarrow a [a_0 \ a \ a_1] [a_0 \ z_0 \ a_1]$$

$$[a_0 \ z_0 \ a_1] \rightarrow a [a_0 \ a \ a_0] [a_0 \ z_0 \ a_1]$$

Turing Machine



2
6
216
68
A

2/2/2
1/1/2
G1/4

7 Tuples

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

Q - Set of States

B - Blank Symbol

Transition Function

$$\delta(q, x) = (p, y, D)$$

$\swarrow \searrow$
 $L \quad R$

p - next state -

y - Tape Symbol being written

Language of M

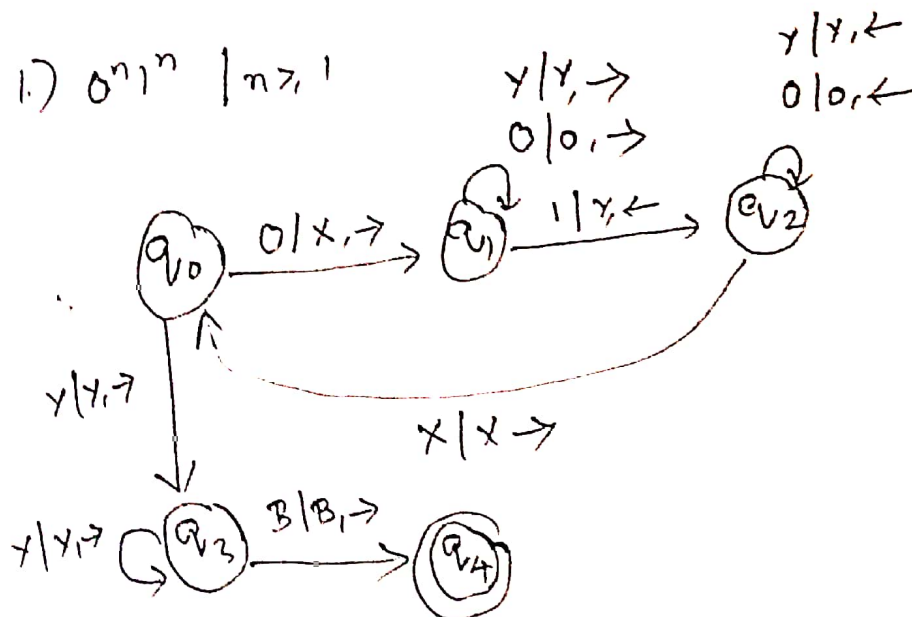
Let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ be a TM,

$$L(M) = \{ w \text{ in } \Sigma^* \mid q_0 w \vdash^* \alpha p \beta^y \}$$

$\alpha \& \beta \Rightarrow$ Tape Strings

$p \Rightarrow$ Final State

1) $0^n 1^n \mid n \geq 1$

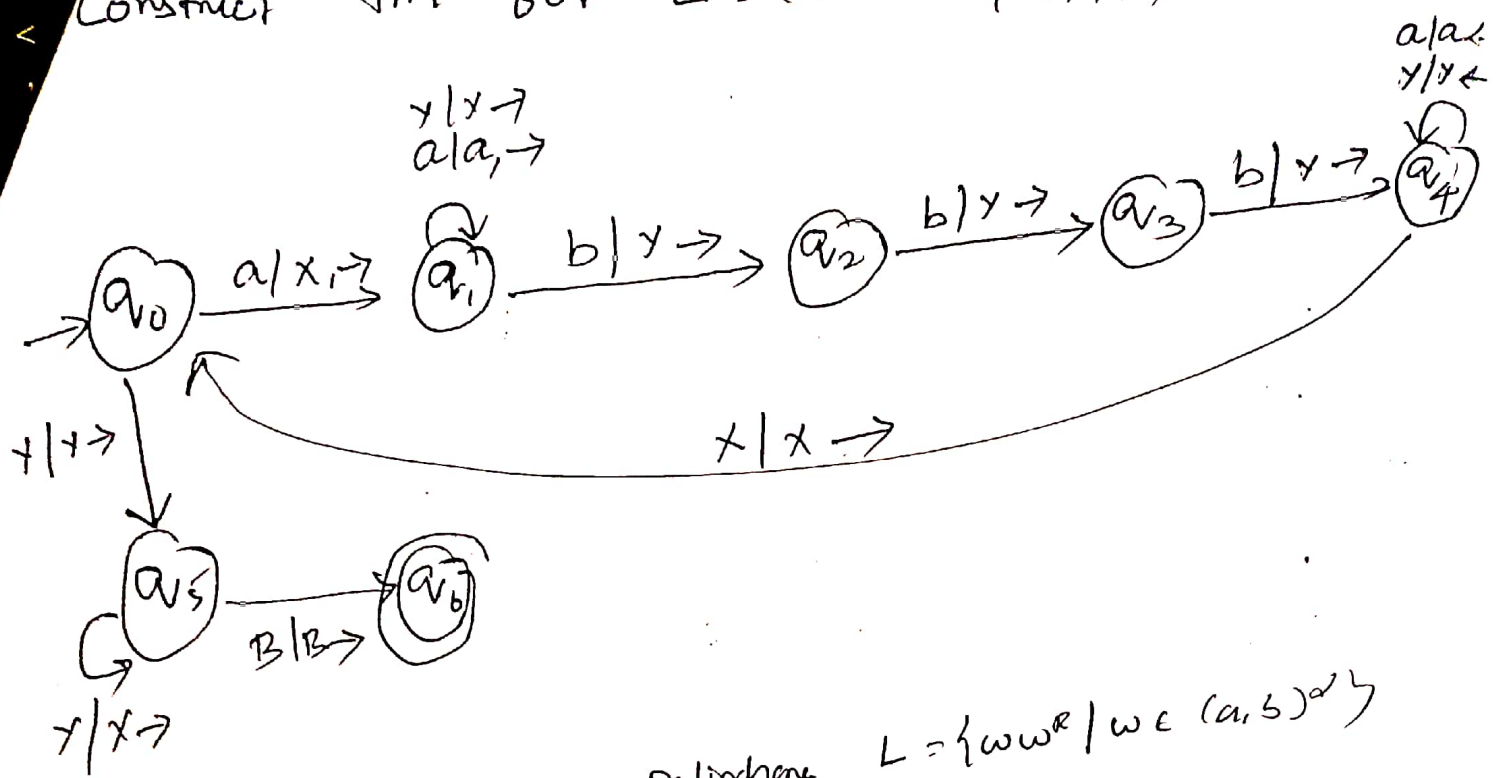


0011
x011
x0x1
xxx1
xxxx

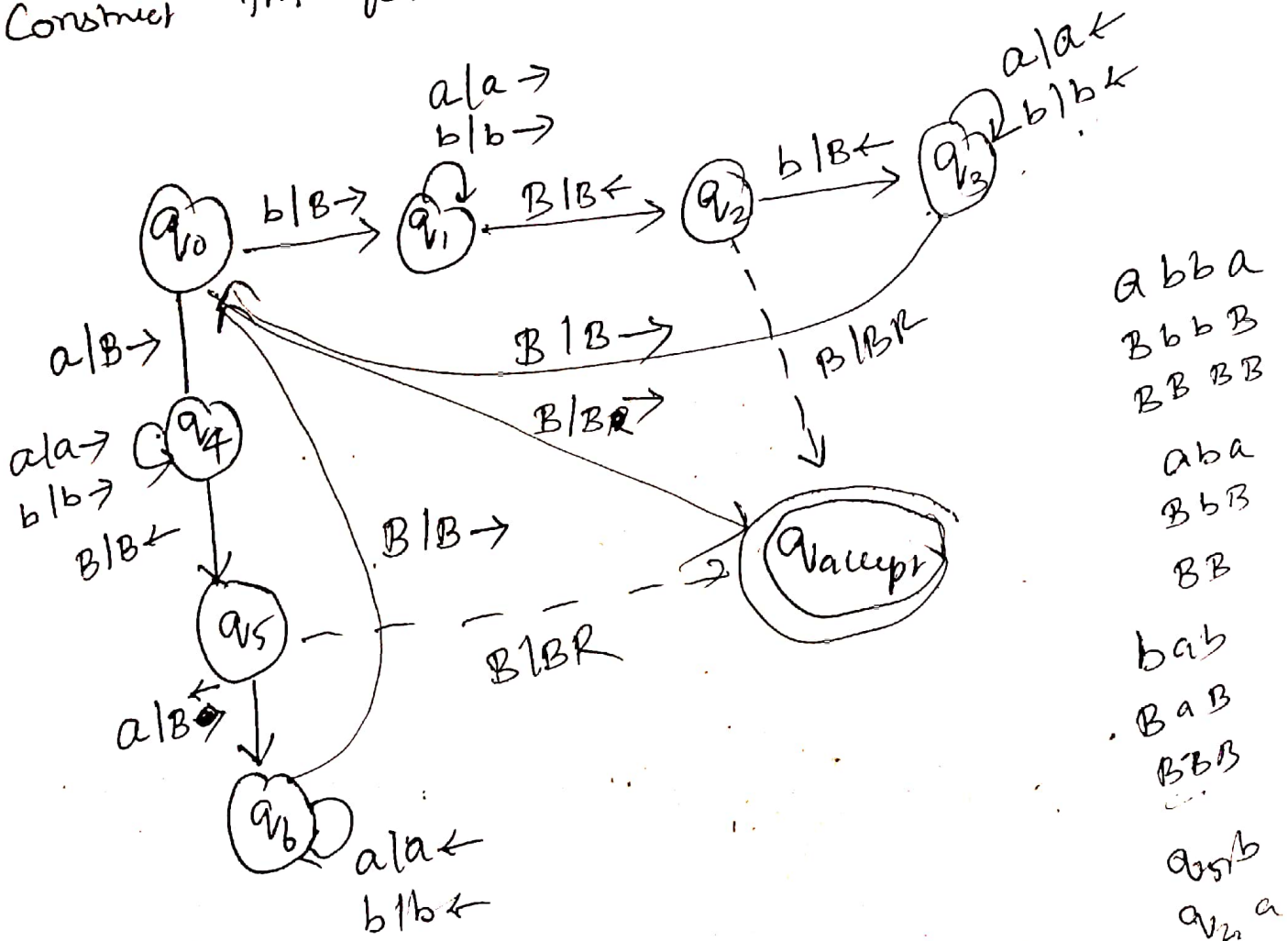
Transition Table

	0	1	B	X	Y
q_0 ($q_1/x \rightarrow$)	-	-	-	-	($q_3/y \rightarrow$)
q_1 ($q_1, 0, \rightarrow$)	(q_2, y, L)	-	-	-	($q_1, y, \rightarrow$)
q_2 ($q_2, 0, \leftarrow$)	-	-	-	($q_0, x, \rightarrow$)	($q_3, y, \leftarrow$)
q_3	-	-	($q_4, x, \rightarrow$)	(q_3)	($q_3, y, \rightarrow$)
q_4	-	-	-	-	-

Construct fm box $L = (a^n b^{3n} \mid n \geq 1)$

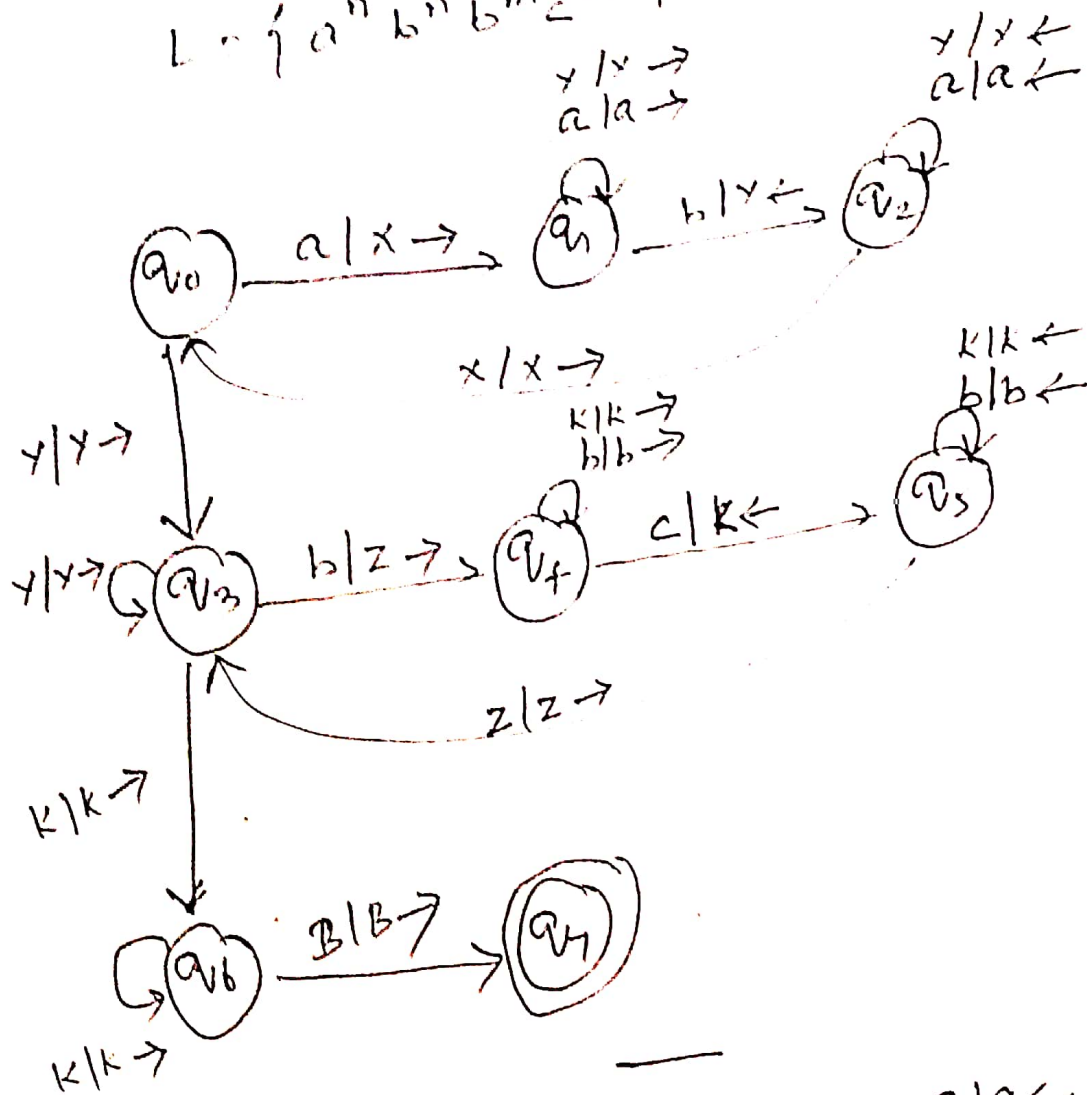


4) Construct gm for palindrone $L = \{ww^R \mid w \in (a,b)^*\}$

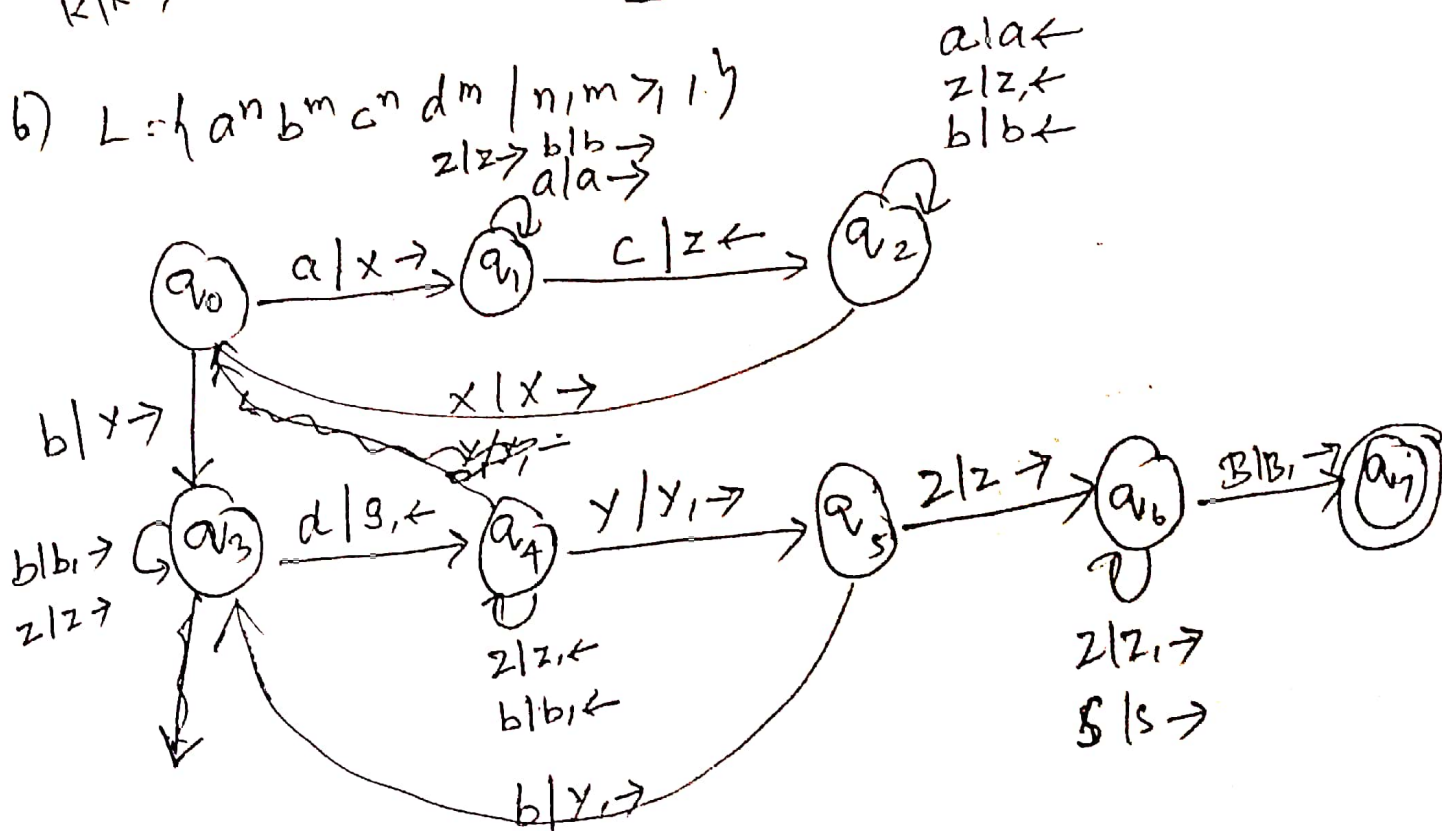


5) $L = \{a^n b^m c^m \mid n, m \geq 1\}$

$L = \{a^n b^n b^m c^m \mid n, m \geq 1\}$



b) $L = \{a^n b^m c^n d^m \mid n, m \geq 1\}$



Recursive Language:

- A Language L is recursive (decidable) if L is the set of strings accepted by some Turing machine that halts on every input.

If $w \in L$, then TM halts in final state

If $w \notin L$, then TM halts in non-final

Recursive Enumerable language.

- A Language L is Recursive Enumerable if L is set of strings accepted by some Turing machine that halts on final state or non-final state or ^{loops} ~~loops~~ forever.

→ Recursive Enumerable Language is called as Turing Recognizable Language

→ Recursive Languages are Turing Decidable Languages

Recursive Languages are also recursive Enumerable

